

## Reference excerpt 01-022

Source: arXiv:1604.05171v1 | License: CC BY 4.0 | Word window: 747-1006

a velocity  $w_T(t)$  that results from a constant component  $w_0$  due to the constant gradient  $\frac{dp}{dz}$  and a time varying component  $w(t)$  due to the time varying  $p_a(t)$ - $\frac{dp}{dz}$ . This system can be described using the Womersley equation that is a linear differential equation where the unknown is the function  $w$  and the pressure gradient  $(\frac{dp(t)}{dz})$  is the source term:  $2w(r, z, t) \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial w(r, z, t)}{\partial r} - w(r, z, t) \frac{\partial}{\partial t} = \frac{1}{z} \frac{dp(t, z)}{\partial z}$  (1) The radial component of the blood velocity and the non-linear terms of the equation are negligible for  $c \gg w$ . [Womersley(1955c)] The solution of Eq.1 and the complete procedure to calculate it are explained in detail in [Sisini et al.(2015b)]. However the solution is there defined up to the multiplicative constant  $c$  and up to the additive constant  $w_0$ . As a consequence, the instantaneous velocity trace, calculated in this way, is proportional but not equal to the actual blood velocity. Pressure waves propagation The pressure is transmitted along the direction AR-IJV according to the following equation  $\frac{dp}{dz} = + \frac{1}{c} \frac{dp}{dt}$  (2) For each cardiac cycle, the pressure in the RA reaches its maximum value between  $t_P$  and  $t_Q$ , where  $t_P$  indicates the ECG P wave and  $t_Q$  indicates the wave Q. In this note, for the sake of simplicity, it is assumed that the pressure in the RA is maximum in correspondence of the instant  $t_P$  which is the average between  $t_P$  and  $t_Q$ .